FUN-4.A.7 The second derivative of a function may determine whether a critical point is the location of a relative (local) maximum or minimum.
FUN-4.A.8 When a continuous function has only one critical point on an interval on its domain and the critical point corresponds to a relative (local) extremum of the function on the interval, then that critical point also corresponds to the absolute (global) extremum of the function on the interval.
FUN-4.D.1 A point on an implicit relation where the first derivative equals zero or does not exist is a critical point of the function.
FUN-4.E.1 Applications of derivatives can be extended to implicitly defined functions.
FUN-4.E.2 Second derivatives involving implicit differentiation may be relations of x, y, and dy/dx.

5-7 & 5-12 The 2nd Derivative Test and Implicit Relations Revisited

Recall: The first derivative test is used to find where any relative extrema occur on a given function.

The First Derivative Test	To find the x values of the relative (local) extrema for a function $f(x)$… • Find the critical values for $f(x)$ - Find where $f'(x) = 0$ or $f'(x)$ does not exist (and $f(x)$ exists) • Create and analyze a sign chart for $f'(x)$ or look at a graph of $f'(x)$ • Find any critical values where $f'(x)$ changes sign ($-$ to $+$ $\rightarrow$ local minimum, $+$ to $-$ $\rightarrow$ 0 local maximum)

The 1st Derivative Test is fairly straightforward and generally quick and easy to use. However, it doesn't always work!

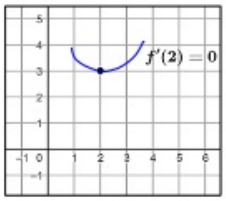
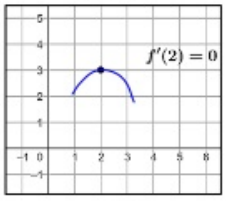
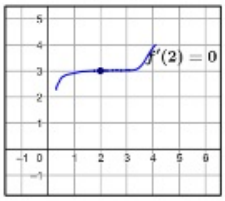
To use the 1st Derivative Test, two things have to be true:

1. We have to be able to find critical values of $f(x)$
2. We have to be able to determine sign change of $f'(x)$

So, what do we do if we cannot use the 1st Derivative Test???
Thankfully, there is another method that can be used to locate relative extrema.

Consider a function $f(x)$ where we are **NOT** given an equation for $f(x)$. However, we do know that $f'(2) = 0$. What does this tell us about $f(x)$? What does this not tell us about $f(x)$?

There are 3 possible cases:

CASE I	CASE II	CASE III
		
At $x = 2$, $f(x)$ has ... Rel min	At $x = 2$, $f(x)$ has ... Rel max	At $x = 2$, $f(x)$ has ... Neither

In the example above, what if we **also** knew that $f''(2) = -5$. What does this tell us about $f(x)$?

$f(x)$ has a horizontal tangent line and is concave down like case II

This is the idea behind the 2nd Derivative Test for relative extrema.

The 2nd Derivative test	
Consider the function $f(x)$ with $f'(c) = 0$. If...	
$f''(c) > 0$	then $f(x)$ has a local **minimum** at $x = c$ by the 2nd Derivative Test
$f''(c) < 0$	then $f(x)$ has a local **maximum** at $x = c$ by the 2nd Derivative Test
$f''(c) = 0$ or $f''(c)$ is undefined	then the 2nd Derivative Test is inconclusive (cannot be applied) **Note:** $f(x)$ **MAY** have a relative minimum, a relative maximum, or neither at $x = c$. The 1st Derivative Test should be applied (if possible)

Important Note: The 2nd Derivative Test is a test for **relative extrema** ... **NOT** points of inflection!

The 1st Derivative Test **AND** the 2nd Derivative Test are **BOTH** for relative extrema. Below is a side by side comparison of how to apply each test.

1st Derivative Test	**2nd Derivative test**
1. Find where $f'(x) = 0$ or undefined. (critical values of f)	1. Find where $f'(x) = 0$. (critical values of f)
2. Use sign chart with f' (or look at graph of f')	2. Plug values where $f'(x) = 0$ into $f''(x)$.
Sample Reasoning/ Explanation:	**Sample Reasoning/ Explanation:**
$f(x)$ has a relative minimum at $x = c$ because $f'(x)$ changes signs from $-$ to $+$ at $x = c$.	$f(x)$ has a relative minimum at $x = c$ because $f'(c) = 0$ and $f''(c) > 0$ (or by 2nd Derivative Test)

When should we use the 2nd Derivative Test???

In most cases, we will use and apply the 1st Derivative Test to find values of x where $f(x)$ has local extrema. However, there are **two** cases that appear on the AP Exam that will require the 2nd Derivative Test:

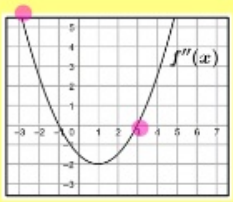
1. When working with a **table of values.**
2. When working with an **implicit curve**.

x	1	2	5	7	9
$f(x)$	4	7	3	-2	1
$f'(x)$	3	0	-1	0	3
$f''(x)$	-6	4	0	-3	2

<u>Example 1:</u> The function f is twice differentiable with selected values of f, f' and f'' given in the table above. Find any relative extrema on the graph of $f(x)$. Explain your reasoning.

$f(2) = 7$ is a relative minimum because $f'(2) = 0$ and $f''(2) = 4 > 0$.

$f(7) = -2$ is a relative maximum because $f'(7) = 0$ and $f''(7) = -3 < 0$.



x	-3	0	2	3	4
$f'(x)$	0	27	5	0	7

Example 2: The function f is twice differentiable for all real values of x and has exactly two critical values. Selected values for f', the derivative of f, are shown in the table above along with a graph of f'', the second derivative of f.

A) At what values of x does $f(x)$ have a critical point?

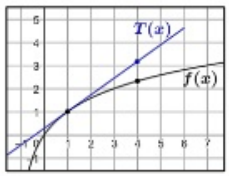
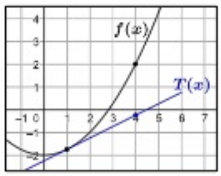
$f(x)$ has a critical point at $x = -3$ and $x = 3$ because $f'(-3) = 0$ and $f'(3) = 0$

B) For each critical point, determine if f has a local extremum at the given x value or state that it cannot be determined.

$f(-3)$ is a local minimum because $f''(-3) > 0$

$f(3)$ cannot be determined because $f''(3) = 0$

Let's revisit Tangent Line Approximations ...

Tangent Line Approximations— Over or Underestimates	
	
$T(4)$ is an **OVERESTIMATE** of $f(4)$ because ... $f(x)$ is concave down on the interval $(0, 6)$	$T(4)$ is an **UNDERESTIMATE** of $f(4)$ because ... $f(x)$ is concave up on the interval $(0, 6)$

AP Exam Tip: When explaining why a tangent line is an overestimate or underestimate, be sure to state that $f(x)$ is concave up/down over an INTERVAL and not just at a point (when possible)!

x	$x < -5$	-5	$-5 < x < -2$	-2	$-2 < x < 3$	3	$3 < x < 11$	11	13	14
$f'(x)$	?	0	−	dne	+	dne	?	3	0	−1
$f''(x)$	−	−2	−	dne	+	dne	−	0	1	?

Example 4: The function $f(x)$ is continuous for all real values of x. The table above includes information about $f'(x)$ and $f''(x)$ for selected values and intervals of x.

A) For $x = -5$ and $x = -2$, determine if the graph of f has a local minimum, local maximum, or neither. Explain your reasoning.

At $x = -5$, the graph of f has a local maximum because $f'(-5) = 0$ and $f''(-5) = -2 < 0$

At $x = -2$, the graph of f has a local minimum because $f'(x)$ changes from negative to positive at $x = -2$

B) Find and classify any additional known values of x where $f(x)$ has a relative extremum. Support your answer.

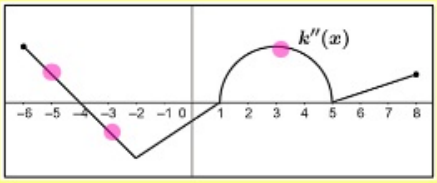
At $x = 13$, the graph of f has a local minimum because $f'(13) = 0$ and $f''(13) = 1 > 0$

C) At which value(s) of x must the graph of f have a point of inflection? Give a reason for your answer.

The graph of f has a point of inflection at $x = -2$ because $f''(x)$ changes from negative to positive at $x = -2$, and a point of inflection at $x = 3$ because $f''(x)$ changes from positive to negative at $x = 3$

D) The line tangent to the graph of $f(x)$ at $x = 6$ is used to approximate $f(10)$. Will this approximation be an over or underestimate of $f(10)$? Explain your reasoning.

The approximation of $f(10)$ will be an overestimate since $f''(x) < 0$ (meaning $f(x)$ is concave down) on the interval $3 < x < 11$.



Example 5: The function $k(x)$ has critical values at $x = -5$, $x = -3$, and $x = 3.693$. The graph of $k''(x)$, the second derivative of $k(x)$, is shown in the figure above.

For which value(s) of x does the graph of $k(x)$ have a local minimum? Give a reason for your answer.

$k(x)$ has a local minimum at $x = -5$ and $x = 3.693$ because these x −values are critical values and $k''(x) > 0$ at these x −values

Example 6: Consider the curve defined by $x^2 + 4y^2 = 7 + 3xy$.

(a) Given that $\frac{dy}{dx} = \frac{3y-2x}{8y-3x}$, show that there is a point P with x-coordinate 3 at which the line tangent to the curve at P is horizontal.

horiz tan line $\rightarrow \frac{dy}{dx} = 0$

$\left.\frac{dy}{dx}\right|_{(3,y)} = \frac{3y-6}{8y-9} = 0$ when $3y-6=0$

$3y=6$

$y=2$

Is $(3,2)$ on the curve?

$3^2+4(2)^2 \stackrel{?}{=} 7+3(3)(2)$

$9+16 \stackrel{?}{=} 7+18$

$25=25$ ✓

$(3,2)$ is on the curve

(b) Given $\frac{d^2y}{dx^2} = \frac{(8y-3x)\left(3\frac{dy}{dx}-2\right)-(3y-2x)\left(8\frac{dy}{dx}-3\right)}{(8y-3x)^2}$, find the value of $\frac{d^2y}{dx^2}$ at the point P in part (a). Does the curve have a local maximum, a local minimum, or neither at the point P? Justify your answer.

$$\left.\frac{d^2y}{dx^2}\right|_{(3,2)} = \frac{(16-9)(0-2)-(6-6)(0-3)}{(16-9)^2}$$

$$= \frac{-14}{49} < 0$$

The curve has a relative max at $(3,2)$ because $\frac{dy}{dx}=0$ and $\frac{d^2y}{dx^2}<0$ at $(3,2)$

Example 9: Let $\boxed{\frac{dy}{dx} = 4x - 3y}$ be the differential equation for the function $y = g(x)$ containing the point (0,1).

(a) Find an expression for $\frac{d^2y}{dx^2}$ in terms of x and y.

$$\frac{d^2y}{dx^2} = 4 - 3\frac{dy}{dx}$$
$$= 4 - 3(4x - 3y)$$
$$= 4 - 12x + 9y$$

(b) Discuss the concavity of the function $y = g(x)$ in Quadrant II.

In QII $x < 0$ and $y > 0$

In QII $\frac{d^2y}{dx^2} = 4 - \underbrace{12x}_{\text{neg}\cdot\text{neg}} + \underbrace{9y}_{\text{pos}\cdot\text{pos}}$ It is the sum of 3 pos. numbers which is pos.

So, In QII $\frac{d^2y}{dx^2} > 0$, and the curve is concave up.